# Water Treatment Plant Digital Twin: FluxForecast

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## Abstract

Data-driven prediction tasks of water quality parameters in water treatment plants (WTPs) pose great opportunities for the development of accurate and cost-effective digital twins, enhancing predictive monitoring, process optimization, and process control. This paper details the initial stages of development of a digital twin for treated water turbidity prediction based on a machine learning model-backbone, focusing on exploratory data analysis and baseline naive and XGBoost models, in advance of the implementation of a previously proposed Wavenet-style temporal convolutional network.

## 1 Introduction

Water treatment plants (WTPs) play a critical role to the functioning of modern societies, transforming raw input water into water fit for human consumption (1). While water consumption applications and standards may vary (i.e. for drinking water, irrigation, manufacturing), strict maintenance of high post-treatment water quality (determined by physical, chemical, and microbial parameters such as turbidity, pH, total coliform) is uncompromisable. Treatment effectiveness also depends on operational parameters (i.e. chemical dosing, flow rates, filtration conditions) and external conditions (i.e. seasonal effectiveness, water source). While many variables associated with water quality are predictable and intuitive, turbidity is multifactorial (2). Turbidity is the measure of relative clarity of water caused by large numbers of suspended particles – making it a good water quality indicator and basis for an anomaly detection algorithm.

There presents a large opportunity in Canada, especially within smaller municipalities, to shift towards the adoption of data-driven methods to assist in managing day-to-day operations, enhancing predictive monitoring and optimization of operational variables. A digital twin based on a machine learning model-backbone can offer competitive maintenance and operational costs, offering potential economic benefit for municipalities. In 2021, the total operating cost of surface water and groundwater treatment plants in Canada was reported to be $1.13 billion Canadian dollars (3). Development of a turbidity prediction model may inform upstream process optimizations for chemical dosing, such that wasteful over-dosing of anti-flocculants and coagulants can be minimized.

A proof-of-concept predictive model of high accuracy leveraging historical and real-time data, capturing complex variable relationships, will greatly support a push towards wider AI-driven model adoption within Canada. A successful model implementation will be deemed such that the standards within the Guidelines for Canadian Drinking Water Quality will be met: the predicted post-filter water turbidity will be within $\pm 0.02$ NTU of the true value - providing equal or better accuracy than turbidimeters (4). While pilot machine learning-based solutions have been tested for water treatment plants in Canada (5), they are typically plant-specific and not widely deployed. In literature, temporal convolutional networks have been implemented in addendum to graph convolutional layers

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or long-short term memory layers in the prediction of various water treatment operational variables (6; 7).

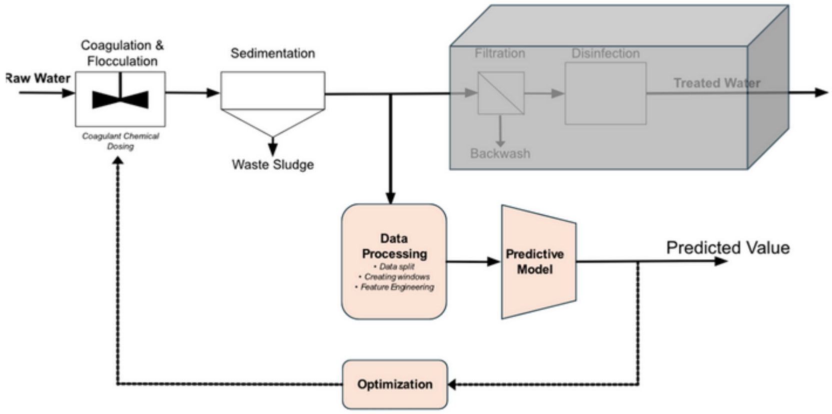


Figure 1: Proposed pipeline for the prediction of treated water turbidity values.

## 2 Exploratory Data Analysis

A dataset from one WTP in Southwestern Ontario with a hydraulic capacity of 125,000 $m^3/d$ is used for turbidity modeling. The data is organized as a daily time series from March $1^{st}$, 2015 to December 31st, 2025, with measurements taken in the raw water, chemical dosing system, filter, and treated water. The raw dataset contains 37 parameters. Treated water sample data and backwash volume data were excluded from analysis as these samples are taken downstream of the filter turbidity, and thus cannot be used in predictive modeling. The dataset used for analysis thus contains 23 parameters and 50,778 non-null values.

Statistical analysis on the dataset was performed to understand the data structure and confirm its quality. A measure of key statistics for the Mean Filter Turbidity parameter is presented in Table 1. Notably, the daily Mean Filter Turbidity is relatively consistent during the historical period, with some peaks. The noted low standard deviation of 0.031 NTU is close to the monitoring equipment noise (i.e., $\pm 0.02$ NTU (4)); therefore, the equipment accuracy may be a significant contributor to the historical variability.

Table 1: Statistic Summary for Mean Filter Turbidity Data

| Statistic | Mean Filter Turbidity |
|---|---|
| Count | 3958 |
| Mean (NTU) | 0.045 |
| Standard Deviation (NTU) | 0.031 |
| Minimum (NTU) | 0.01 |
| 25th Percentile (NTU) | 0.03 |
| 50th Percentile (NTU) | 0.04 |
| 75th Percentile (NTU) | 0.05 |
| 90th Percentile (NTU) | 0.07 |
| Maximum (NTU) | 0.77 |

The correlation coefficient between the Mean Filter Turbidity and other parameters were determined to confirm the data is plausible with theoretical WTP knowledge and to predict the most impactful model features. Table 2 depicts the parameters with the top 5 highest positive and negative correlations with the Mean Filter Turbidity. The correlations overall align with expected trends: turbidity increases during high flow and seasonally (i.e., high water temperature, higher total organic carbon). However, negative correlations with raw water nitrate and ammonia are unexpected, as increases in these parameters typically follow algal blooms, which should cause a turbidity increase.

Table 2: Top Correlations with Mean Filter Turbidity

| Top Positive Correlation Parameters | Correlation | Top Negative Correlation Parameters | Correlation |
|---|---|---|---|
| Raw Water Temperature | 0.209 | Raw Water Nitrate | -0.235 |
| Max Filter Turbidity | 0.166 | Total Filter Runtime | -0.106 |
| Raw Water Flow | 0.154 | Raw Water Total Ammonia | -0.069 |
| Raw Water Total Organic Carbon | 0.134 | Raw Water Dissolved Organic Carbon | -0.060 |
| Raw Water Alkalinity | 0.085 | Raw Water PH | -0.043 |

The data was pre-processed to remove all null values via forward filling. Note that forward filling was performed in the test and train sets independently to prevent data leakage. A two-sample Kolmogorov-Smirnov (K-S) test was performed on each parameter before and after preprocessing. For all parameters the null hypothesis was not rejected, thus preprocessing likely did not alter the data distribution. The K-S test was also performed to compare the train set and test set. Based on the K-S statistic, the data distribution was different in each set for most parameters. This can be explained by the nature of the time series data, as small changes in service population, climate, and source water quality are expected year over year. However, this presents a challenge for the model: differing data distributions may reduce prediction accuracy.

## 3 Data Processing

Our project utilizes tabulated operational data from a Water Treatment Plant (WTP) located in Eastern Ontario. The dataset is structured as a daily time series. The data represents sampling of the flow and operational measurements from different stages of the treatment process, including chemical dosage, temperature, pH, turbidity, and other water-quality parameters.

The original excel/csv dataset spans 2006-2025 and contains 42 initial variables. For data reliability, the analysis was restricted to 2015-2025, resulting in 3,958 daily observations. The focus of this project is to predict treated water (TW) quality parameters using raw water and operational variables as model inputs. The dataset presents several practical challenges common to real-world industrial data. Mainly, variables are sampled at irregular frequencies, with certain laboratory and water-quality measurements recorded intermittently rather than daily, leading to frequent gaps in the data. Moreover, water treatment plant data exhibits strong temporal dependence, requiring modeling approaches that preserve this temporal feature of our data (8). These challenges required us to create a structured preprocessing pipeline to ensure data quality, and usability.

Data processing was based on four primary pillars: data cleaning, feature engineering, sliding-window construction, and time-based data splitting. The dataset was truncated to begin in 2015, and only relevant upstream variables (raw water and operational measurements) were retained. Downstream variables were excluded to avoid target leakage. All features were ensured to be numerical (e.g., convert cyclical date/time variables (hours, days, months) into numerical using cos/sin), and anomalous sensor readings were discovered during our EDA.

Because of the time-series nature of the data, random splitting was avoided. Instead, the dataset was partitioned by calendar year to preserve seasonal structure and prevent temporal leakage. The training set includes data from 2015-2022, the validation set corresponds to 2023, and the test set includes 2024-2025. This strategy maintains yearly and seasonal behavior across splits, although it assumes relative consistency in system behavior between years.

Because several variables are measured intermittently and to preserve temporal information without artificially interpolating values, we created two additional engineered features, for each original feature $x(t)$: a mask feature and a previously measured feature. The mask variable is a binary indicator equal to 1 when the variable is observed at time t, and 0 otherwise. The previously measured value represents the most recent observed measurement carried forward in time. This design allows the model to distinguish between directly observed and imputed values while capturing the "staleness" of measurements (8; 9).

Forward filling was applied to prevent computational errors when feeding the data into neural network models (e.g., NaNs). However, the mask variables ensure that the model remains "aware" of missingness and avoids learning misleading patterns from imputed values. Given the strong temporal dependence in WTP performance, fixed-length sliding windows were constructed to capture dynamic system behavior. For a window size $w$, each input sample consists of the sequence of observations from time $t - w + 1$ to time $t$ (6). The associated target corresponds to the treated water parameter at the desired prediction horizon. This windowing approach enables the model to learn time-lagged dependencies, cumulative operational effects, and enabling the representation of short-term system "memory". The model output is aligned with the final time step of each window, ensuring consistent temporal mapping between inputs and targets. The window and shift size are treated as hyperparameters.

For each constructed window, a corresponding treated water parameter value (e.g., turbidity or total aluminum) is assigned as the prediction target. The resulting dataset is therefore structured as a collection of input-output pairs (and array of tuples `[(window_df, target), ...]`), where each input is a matrix of dimension $w \times d$ (window length by number of features) and each output is an integer target value. Each data split (training, validation, and testing) consists of an array of window-target tuples.

For a window size of seven days and a shift size of one day, each window contains seven time steps and 90 engineered features. Under this configuration, the final dataset consists of 2,856 training windows, 358 validation windows, and 724 testing windows. Future edits to the preprocessing pipeline will depend on the specific modeling approach employed. Additional steps might include backward filling and feature normalization or scaling, particularly for neural network-based models that are sensitive to feature magnitude. Model performance will also be evaluated under different sliding window sizes and shift steps. The optimal configuration will depend on the physical behavior of the target variable, as some treated water parameters are influenced more strongly by recent operational conditions, while others exhibit longer memory effects. The overall data processing pipeline is sequential: data splitting, data preprocessing (including cleaning and feature engineering), and sliding window construction.

## 4 Modelling Decisions

Because XGBoost requires tabular (2D) input rather than sequential tensors, each window matrix (`window_size x number_of_features`) was flattened into a single one-dimensional feature vector by concatenating all feature values across the time steps. This effectively converted temporal sequences into lagged features, allowing the model to learn autoregressive and cross-feature interactions within the window.

To make inferences on ideal temporal window sizes, the effect of how a raw water turbidity measurement on one day influences its value on subsequent days was visualized via autocorrelation and partial autocorrelation plots. The autocorrelative accounts for intermediate day values, whereas partial autocorrelation measures the direct correlation and effect between two compared days. Based on the autocorrelation function, a window size between 7-14 days can be hypothesized to be most effective in capturing local changes in mean raw water turbidity values, given that the correlation decreases towards a plateau of $\sim 0.25$. A prediction timeframe of 1 day after the training window is hypothetically most accurate for filtered water turbidity prediction given the sharp dropoff in partial autocorrelation beyond 1 day.

Practically, a model training on a smaller window size would be more useful, allowing for iterative process monitoring with short time windows. Exploratory data analysis reveals that window size matters very little when the shift step of 1 is utilized: given that most data is overlapping, supporting the behaviour exhibited via autoregressive plot analysis. Conversely, the shift size increase will lead to less overlapping fitting data, yielding volatile naive $R^2$ values.

## 5 Results and Discussion

Based on our results we can see that turbidity forecasting shows a distinct behavior: strong short-term persistence, rapid decay of predictability. The experimental results indicate that improved accuracy in mean turbidity forecasting is fundamentally constrained by the inherent noise and variance of the target feature (variance 0.0015). For +1 day prediction of mean turbidity, the naïve baseline (predicting yesterday's value) already performed very strongly, with Test $R^2 \approx 0.74$. Our XGBoost model using a 7-day window and step $= 7$ improved this to about Test $R^2 \approx 0.78$, with Validation

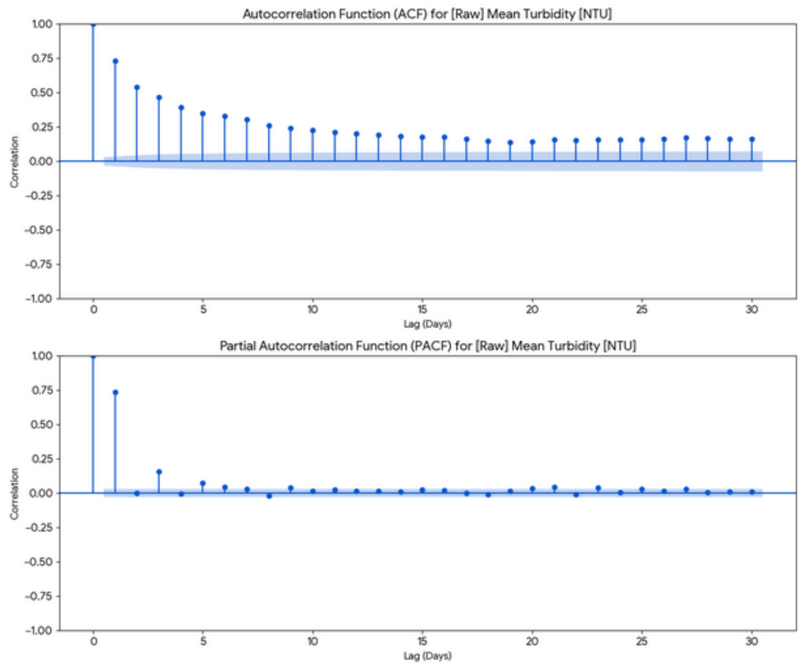


Figure 2: Correlation and autocorrelation function plots for mean turbidity of raw water.

$R^2 \approx 0.90$ and good generalization (Train $R^2 \approx 0.92$). This indicates that the most predictable signals are autoregressive and that the model extracts only modest additional structure beyond persistence.

When we varied window size and shift, we observed that performance was highly sensitive to sampling. Changing the shift (e.g., step $= 7$ vs step $= 1$) also changed which timestamps were evaluated. Step $= 7$ produces non-overlapping windows and effectively samples every 7th day, often yielding more stable and independent training samples. Step $= 1$ introduces highly overlapping windows, increasing redundancy and exposure to high-frequency noise, which sometimes led to poorer test performance. Window size influences performance by controlling how much short-term decay information is captured, while shift affects statistical sampling and redundancy.

Statistically, turbidity is a difficult variable. For mean turbidity at +1 day, autocorrelation is high, making it predictable. But at +3 days, naïve performance dropped sharply to $R^2 \approx 0.24$, and even the model achieved only $\approx 0.24$, showing that predictability decays quickly. Hyperparameter optimization on the baseline XGBoost model was performed using Bayesian optimization. The Optimized XGBoost model achieved Train, Validation, and Test $R^2$ values of 0.86, 0.84, and 0.66, respectively. Optimized XGBoost arguments are presented in Appendix A. This model achieves a MSE of 0.000213, a MAE of 0.00832, and Max Error of 0.0719.

## 6 Future Prospects

Appropriate Wavenet-style temporal convolutional network architectures have been identified, and its results will be one of the next goals. The baseline model using XGBoost achieves the success condition of the project proposal on average: the mean absolute error of the model is within the accuracy range of turbidimeters. However, the model does not accurately predict high turbidity events, as evidenced by the large max error and low $R^2$. Additionally, high model instability is experienced.

As noted, the Mean Filter Turbidity exhibited a low standard deviation of 0.031 NTU relative to its mean of 0.045 NTU. Highly clustered data provided limited discriminatory power for predictive modelling. Additionally, the standard deviation is close to the turbidimeter noise of $\pm 0.02$ NTU (4). Turbidimeter noise was likely a higher contributor to changes in the Mean Filter Turbidity when compared to other model features. As a result, the current data is unlikely to provide a robust model for predicting Mean Filter Turbidity.

Alternative data handling techniques will be explored. One strategy to improve prediction may be to use a shorter data range. This method would increase the similarity between the test and train data, as less compounding effects (population, climate, and raw water change) would alter the test and train data distributions. However, this strategy does address the challenge in modelling the low variance data. Thus, a second strategy is to model an alternative parameter, such as the treated water aluminum concentration (Table 3). Changing the target value to one with a higher variance would facilitate modelling and still provide a prediction on the treated water quality.

A third strategy would pivot our project from predicting the exact next-day value, to exceedance or event prediction. For example, the model could estimate the probability that turbidity exceeds a regulatory or operational threshold within the next 1-3 days. The fourth strategy is to shift forecasting objectives from point prediction to interval or range prediction (e.g., predicting upper and lower bounds or quantiles). The last two strategies would necessitate a change in scope for the project. We will develop fast experiments to determine what would be the best course of action.

Table 3: Statistic Comparison for Mean Filter Turbidity and Treated Water Aluminum Data

| Statistic | Mean Filter Turbidity (NTU) | Treated Water Aluminum (mg/L) |
|---|---|---|
| Count (-) | 3958 | 3954 |
| Mean | 0.045 | 0.073 |
| Standard Deviation | 0.031 | 0.065 |
| Minimum | 0.01 | 0.00 |
| 25th Percentile | 0.03 | 0.03 |
| 50th Percentile | 0.04 | 0.059 |
| 75th Percentile | 0.05 | 0.092 |
| 90th Percentile | 0.07 | 0.137 |
| Maximum | 0.77 | 0.97 |

## 7 Repository

Open-access to the repository may be achieved at https://github.com/kevzhu14/FluxForecast–Water-Treatment-Plant-Turbidity-Predictor—Anomaly-Detection

## References

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# A Optimized XGBoost Hyperparameters

Table 4: Optimized XGBoost Hyperparameters Used

| Hyperparameter | Value |
|---|---|
| colsample_bytree | 0.9716372546447916 |
| gamma | 0.0004901288721936043 |
| learning_rate | 0.03969052651907344 |
| max_depth | 6 |
| min_child_weight | 3 |
| n_estimators | 280 |
| reg_alpha | 0.5366946896712355 |
| reg_lambda | 14.119219963648737 |
| subsample | 0.6427084493906863 |
| early_stopping_rounds | 15 |

All other parameters were set as the XGBoost baseline values.